

A Survey of Federal and State Incentives for Household Adoption of Beneficial Technologies

Prepared June 2026. This survey documents the incentive instruments that U.S. federal and state governments deploy to accelerate household adoption of durable, socially beneficial technologies, and codes each instrument against the four structural features of a forward-looking durable-goods adoption model. The object is to establish that the mechanism estimated for rooftop solar generalizes: the same four features recur across the broader incentive landscape, which licenses the model's application to electric vehicles, heat pumps, and other incentivized household technologies.

1. Scope and method

The survey covers sixteen federal programs and 205 state and major-utility programs across all fifty states, spanning rooftop solar, battery storage, electric vehicles and chargers, air-source heat pumps and heat pump water heaters, weatherization, efficient appliances, electrical-panel upgrades, water efficiency, broadband, and demand-response devices. Program identification draws on the Database of State Incentives for Renewables and Efficiency (DSIRE), administering-agency program pages, and the U.S. Department of Energy and Internal Revenue Service guidance current to June 2026. State coverage captures each jurisdiction's flagship statewide and large-utility programs by technology category; it is representative of the regime rather than an exhaustive enumeration of every municipal line item. The complete machine-readable matrix, with dollar amounts, eligibility, current status, source links, and the feature codes discussed below, accompanies this document as a spreadsheet.

2. A four-feature lens

Each instrument is coded on four features. The codes are Y when a feature is present, Partial when it is present but attenuated, and No when it is absent.

NPV durable good. The incentive targets a durable asset whose adoption rests on the net present value of a future benefit stream weighed against an upfront cost. Solar, storage, vehicles, heat pumps, weatherization, and panel upgrades satisfy this feature directly; appliances and water-efficiency retrofits satisfy it partially; broadband, a recurring service, does not. Borenstein (2017) formalizes the private-NPV calculus for residential solar, and the same discounted-cash-flow logic governs the vehicle and heat-pump decisions.

Time-varying incentive path. The incentive amount moves over time through a scheduled step-down, a statutory sunset, or budget-driven exhaustion. Block-declining production tariffs (the Massachusetts SMART program, New York NY-Sun, Illinois Shines, Rhode Island REG) embody scheduled declines; the federal credit sunsets embody statutory ones; first-come rebate budgets that reach waitlist status embody administrative ones. Hughes and Podolefsky (2015) show that the California Solar Initiative's declining block structure shaped the timing and volume of adoption, which is the identifying variation this survey documents at scale.

Visibility and peer effects. The adopted technology is observable to neighbors, generating the visibility channel through which adoption propagates. Rooftop solar, electric vehicles, and turf-replacement landscaping are visible; insulation, water heaters, panels, and broadband are not. Bollinger and Gillingham (2012) estimate sizable peer effects in solar diffusion, and Bollinger et al. (2022) extend the visibility mechanism across durable goods.

Forward-looking timing. The incentive creates an intertemporal reason to time adoption, through announced deadlines, scheduled declines, or act-before-funds-are-gone budget constraints. De Groote and Verboven (2019) show that solar adopters respond to the present value of future subsidy streams rather than to instantaneous cash flow, evidence that households optimize dynamically. Sallee (2011) documents the same forward-looking timing around vehicle tax-credit phase-outs.

3. The federal layer after PL 119-21

The federal incentive structure changed decisively in the past year. The One Big Beautiful Bill Act (Public Law 119-21, July 4, 2025) terminated the Section 25D Residential Clean Energy Credit and the Section 25C Energy Efficient Home Improvement Credit for property placed in service after December 31, 2025, and terminated the Section 30D and Section 25E vehicle credits for vehicles acquired after September 30, 2025. The Section 30C charger credit remains available only for property placed in service through June 30, 2026. The Inflation Reduction Act rebate programs (HEAR and HOMES) survive as state-administered point-of-sale rebates, and the income-targeted backstops (the Weatherization Assistance Program, LIHEAP, and Lifeline) remain funded through fiscal year 2026. Table 1 summarizes the federal layer.

Table 1. Principal federal household-technology incentives, status as of June 2026.

Federal program	Amount	Mechanism	Status (June 2026)
Sec. 25D Residential Clean Energy (solar, storage, geothermal)	30% of cost, uncapped	Tax credit	TERMINATED after 12/31/2025
Sec. 25C Energy Efficient Home Improvement (heat pumps, insulation)	30%, up to \$1,200/yr (+ \$2,000 heat pump)	Tax credit	TERMINATED after 12/31/2025
Sec. 30D New Clean Vehicle	Up to \$7,500	Tax credit	TERMINATED after 9/30/2025
Sec. 25E Previously-Owned Clean Vehicle	Up to \$4,000	Tax credit	TERMINATED after 9/30/2025
Sec. 30C Refueling Property (EV charger)	30%, up to \$1,000	Tax credit	Sunsets 6/30/2026
HEAR rebates (IRA Sec. 50122)	Up to \$14,000/household	Point-of-sale rebate	Active, state-administered
HOMES rebates (IRA Sec. 50121)	Up to \$8,000	Performance rebate	Active, state-administered
Weatherization Assistance Program	~\$6,500/unit avg	Grant	Active (FY26 ~\$329M)
LIHEAP	Varies	Bill subsidy / grant	Active (~\$4.05B FY26)
Lifeline	\$9.25/mo	Bill subsidy	Active
Affordable Connectivity Program	\$30/mo	Bill subsidy	TERMINATED (2024)

These sunsets are not a footnote to the analysis; they are a vivid demonstration of features two and four. A credit with a known termination date is a time-varying incentive path that rewards forward-looking households for pulling adoption forward, exactly the intertemporal margin the model is built to capture. The expiration of the thirty-percent solar credit at the close of 2025 raises the relative weight of the state programs catalogued here, which now constitute the primary subsidy for most households.

4. The state layer

State regimes stratify sharply. A leading tier (California, New York, Massachusetts, New Jersey, Rhode Island, Connecticut, Colorado, New Mexico, Oregon, Washington, Illinois, Minnesota, Maryland) combines statewide tax credits or production tariffs with stacked rebates for storage, heat pumps, and vehicles. A minimal tier (Wyoming, Idaho, the Dakotas, and much of the Southeast) offers little beyond net metering, property-tax exemptions, and federally funded weatherization. This cross-sectional dispersion in incentive generosity is itself useful: it is the policy variation that identifies adoption responses, and it confirms that the instruments the model treats as drivers are present in materially different doses across jurisdictions.

Six mechanism families recur. Investment tax credits and deductions (twenty programs) reduce upfront cost against the buyer's tax liability. Sales- and property-tax exemptions (nineteen programs) are the most stable instrument, with no scheduled decline. Upfront rebates (the largest family, 107 programs) are budget-administered and therefore subject to waitlists and exhaustion. Production and performance payments (twenty-one programs), including net-metering successor tariffs and battery demand-response credits, pay per unit of output over a multi-year horizon. Low-interest loans and on-bill financing (eight programs) address liquidity. Bill subsidies (broadband and energy assistance) address recurring service cost. The mechanism a state chooses determines which of the four features is most pronounced, and production tariffs activate all four at once.

5. Generalizability: the pattern across technologies

Table 2 codes the four features by technology category and reports the number of catalogued programs in each. The pattern supports the model's external reach. Rooftop solar, the estimated setting, scores Y on all four features. Electric vehicles score Y on all four as well: they are durable NPV goods, they are socially visible, their federal credit carried a hard deadline, and several state credits decline on an announced schedule. Heat pumps and storage share the durable-NPV and time-path features and differ mainly in visibility. The instruments that govern these technologies are the same instruments that govern solar, administered by the same agencies under the same budget cycles. Twenty-one of the catalogued programs, concentrated in rooftop solar production tariffs with scheduled declines, score Y on all four features simultaneously, which marks them as the closest empirical analogues to the estimated model.

Table 2. Model-feature coding by technology category, with program counts (federal + state).

Technology	NPV good	Time path	Visibility	Timing	Programs	Note
Rooftop solar PV	Y	Y	Y	Y	89	Closest analogue to the estimated solar model
Battery storage	Y	Partial	Partial	Partial	22	Often co-installed; performance/DR tariffs add a path
Electric vehicles	Y	Y	Y	Y	19	Durable, visible, deadline-driven (federal sunset 9/2025)
EV chargers	Partial	Y	Partial	Y	9	Enabling durable; 30C deadline 6/2026
Heat pumps (ASHP)	Y	Partial	Partial	Partial	56	Durable NPV good; budget-cycled rebates

Technology	NPV good	Time path	Visibility	Timing	Programs	Note
Heat pump water heaters	Y	Partial	No	Partial	19	Durable but low visibility
Weatherization / insulation	Y	Partial	No	Partial	38	Durable, invisible; mostly income-targeted
Efficient appliances	Partial	Partial	No	Partial	31	Shorter-lived; rebate-cycled
Electric panel / wiring	Y	Partial	No	Partial	17	Enabling durable upgrade
Water-efficiency / turf	Partial	Partial	Partial	Partial	5	Turf removal is visible and durable
Broadband adoption	No	Partial	No	Partial	5	Service subsidy, not a durable asset

The reading is that the visibility feature is the binding restriction on generalization, not the NPV or timing features. Where a technology is durable and incentivized on a declining or deadline-bound path but is not socially observable (heat pump water heaters, insulation, panels), three of the four features transfer and the peer-effect channel attenuates. Where a technology is durable, incentivized, and visible (vehicles, exterior heat pumps, turf conversion), all four transfer. This delineates the model's domain of application precisely and answers the scope question directly: the framework applies wherever a durable household technology is subsidized on a time-varying path, with the visibility channel scaling with the observability of the installed good.

6. Caveats

Three qualifications bound the survey. First, dollar amounts and program statuses change with program years and budget cycles; the linked source governs any individual figure. Second, the feature codes for the NPV and visibility dimensions are assigned by technology and the codes for the time-path and timing dimensions are assigned from mechanism and current status; they are analyst codings for pattern-matching, not attributes claimed by administrators. Third, state coverage is representative at the flagship level and does not enumerate every municipal water-utility rebate or cooperative-utility program, which number in the thousands in DSIRE.

References

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